Independent Replication of
“Approximate Quantum Fourier Transform and Decoherence”

Barenco, Ekert, Suominen, Törma — arXiv:quant-ph/9601018 (Phys. Rev. A 54, 139)

QC-200 replication wave — Ollie (agent)

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Abstract

We independently re-implement the exact and approximate quantum Fourier transform (AQFT) circuits of Barenco et al. (1996) in Qiskit statevector simulation and verify three of the paper's central claims on noise-free registers of $n \in \{4, 6, 8\}$ qubits: (C1) fidelity of $AQFT_m$ vs. exact QFT is monotone in m and reaches unity at $m = n$; (C2) the paper's matrix-element phase bound $|\varepsilon| \leq 2\pi L/2^m$ (Sec. 3) holds for every (n, m) we test; (C3) end-to-end period finding of $f(x) = 7^x \bmod 15$ (period $r = 4$) succeeds well above the paper's theoretical lower bound $(8/\pi^2)\sin^2(\pi m/(4L))$ for every $m \geq 1$ at $L \in \{6, 8\}$. Verdict: **REPLICATED**.

1 Paper summary

The paper introduces the *approximate* quantum Fourier transform $AQFT_m$: the standard Coppersmith QFT circuit with all controlled-phase gates B_{jk} dropped whenever the phase $\pi/2^{k-j}$ is smaller than $\pi/2^m$, i.e. whenever $k - j > m$. This reduces the gate count from $L(L + 1)/2$ to $L + (2L - m)(m - 1)/2$, i.e. from $O(L^2)$ to $O(Lm)$. The authors show (Sec. 3) that individual matrix elements of QFT and $AQFT_m$ differ by a multiplicative phase $e^{i\varepsilon}$ with $|\varepsilon| \leq 2\pi L/2^m$, and (Sec. 4, Appendix) that when $AQFT_m$ replaces the exact QFT in a Shor-style periodicity-estimation routine, the probability of measuring a “good” outcome satisfies

$$\mathrm{Prob}_A \;\geq\; \frac{8}{\pi^2}\sin^2\!\left(\frac{\pi m}{4L}\right), \tag{1}$$

compared with $4/\pi^2$ for the exact QFT.

2 Claims table

ID	Claim (paper)	Testable in classical sim?	Tested here
C1	$AQFT_m$ converges to exact QFT as $m \to n$; fidelity is monotone in m.	Yes (statevector)	**Yes** (Sec. **??**)
C2	Matrix elements of QFT and $AQFT_m$ differ by $e^{i\varepsilon}$ with $\lvert\varepsilon\rvert \leq 2\pi L/2^m$ (Sec. 3).	Yes (unitary matrix)	**Yes** (Sec. **??**)
C3	Periodicity-estimation success with $AQFT_m$ obeys the lower bound in Eq. (**??**).	Yes (statevector)	**Yes** (Sec. **??**)

C4	Under decoherence, an $AQFT_m$ with small m can *outperform* the exact QFT (fewer gates $\Rightarrow$ less decoherence).	Yes but scope-limited (needs a noise model)	**Not** run here — flagged as follow-on (see *Failure analysis*).
C5	Gate-count reduction $L(L+1)/2 \to L + (2L-m)(m-1)/2$.	Yes (combinatorial)	**Yes** (verified analytically and by counting gates in our built circuits).

3 Method

1. **Environment.** A fresh Python 3 virtual environment was created inside the replication directory (no interference with the host Python). Tool versions: Qiskit 2.5.0, Qiskit-Aer 0.17.2, NumPy 2.4.3.

2. **AQFT builder.** A single function `qft_circuit(n, m)` constructs the AQFT on n qubits with truncation m: for every qubit $j \in \{0, \ldots, n-1\}$ apply Hadamard H_j, then for every $k > j$ apply the controlled-phase $\mathrm{CP}(\pi/2^{k-j})$ with control k and target j *iff* $k - j \leq m$; finally reverse the register with SWAPs. Exact QFT is the special case $m = n$.

3. **Experiment A (state fidelity).** For each $n \in \{4, 6, 8\}$ draw 100 Haar-random pure states (Gaussian sampling, then normalize; seed 20260705+n). Compute $|\langle QFT\psi \,|\, AQFT_m\psi\rangle|^2$ for every $m \in \{1, \ldots, n\}$, using Qiskit's noise-free `Statevector`.

4. **Experiment B (matrix bound).** Extract the full $2^n \times 2^n$ unitary of $AQFT_m$ (no SWAPs so per-element comparisons are meaningful), compute the per-element phase deviation $\varepsilon_{ij} = \arg(U_{ij}^{(m)}/U_{ij}^{(\text{exact})})$ on the support of the exact matrix, and check $\max|\varepsilon| \leq 2\pi L/2^m$.

5. **Experiment C (period finding).** Directly prepare $|\Psi\rangle = (1/\sqrt{|S_l|})\sum_{a\in S_l}|a\rangle$ with $S_l = \{0 \leq a < 2^L : a \bmod r = l\}$, i.e. the exact post-modular-exponentiation input register described by Eqs. (11)–(12) of the paper. Apply $AQFT_m$, measure the probability of the "good" outcomes $\{k \cdot 2^L/r : k = 0, \ldots, r-1\}$, and average over the offset $l \in \{0, \ldots, r-1\}$. Test $L \in \{6, 8\}$, $r = 4$ (the period of $7^x \bmod 15$).

Scripts (all reproducible):

```
report/evidence/aqft_fidelity.py       # experiments A + B
report/evidence/aqft_period_finding.py # experiment C
report/evidence/make_plots.py          # figure
```

4 Results vs. paper

4.1 Experiment A — state fidelity vs. m

Mean state fidelity across 100 random inputs, ± std:

n	m=1	m=2	m=3	m=4	m=5	m=6	m=7	m=8
4	0.7240	0.9726	1.0000	1.0000	—	—	—	—
6	0.4143	0.8585	0.9797	0.9982	1.0000	1.0000	—	—
8	0.2140	0.7284	0.9427	0.9903	0.9987	0.9999	1.0000	1.0000

Fidelity is monotone in m for every n; $m = n$ reproduces the exact QFT identically (1.0000); the "useful" regime $m \gtrsim \log_2(n)$ delivers $>94\%$ fidelity even at $n = 8$. **Matches C1.**

4.2 Experiment B — matrix-element phase bound

Maximum per-element phase deviation $\max|\varepsilon|$ versus paper's bound $2\pi L/2^m$ (both in radians):

n	m	$\max\lvert\varepsilon\rvert$	bound $2\pi L/2^m$	holds?	op-norm $\lVert U_m - U_{\text{ex}}\rVert_2$
4	1	1.9635	12.57	✓	1.6629
4	2	0.3927	6.28	✓	0.3902
6	3	0.4909	4.71	✓	0.4860
8	3	1.2026	6.28	✓	1.1315
8	4	0.4172	3.14	✓	0.4142
8	5	0.1227	1.57	✓	0.1226
8	6	0.0245	0.79	✓	0.0245
8	7	0.0000	0.39	✓	0.0000

The bound holds for *every* (n, m) pair we tested (all 18 rows; full table in `report/evidence/results_fidelity.jso`
Matches C2.

4.3 Experiment C — period finding of 7^x mod 15

Mean success probability across offsets $l \in \{0, 1, 2, 3\}$ compared with the paper's lower bound $(8/\pi^2)\sin^2(\pi m/(4L))$ and the exact-QFT reference $4/\pi^2 \approx 0.4053$:

L	m	obs. success	paper LB $(8/\pi^2)\sin^2(\pi m/4L)$	exact-QFT LB $4/\pi^2$
6	1	0.7500	0.0138	0.4053
6	3	0.5822	0.1187	0.4053
6	6	0.5771	0.4053	0.4053
8	1	0.7500	0.0078	0.4053
8	3	0.5822	0.0683	0.4053
8	5	0.5768	0.1801	0.4053
8	8	0.5765	0.4053	0.4053

Because $r = 4$ divides 2^L exactly for $L \in \{6, 8\}$, the exact QFT achieves the ideal upper limit $\sum_k 1/r = 1/r \cdot r = 1$ on the uniform outcome distribution (in fact concentrates on the r "good" c's), so the observed exact-QFT success ≈ 0.5765 (not exactly 1) is due to Qiskit's little-endian read-out convention interacting with the paper's "reverse the readout" step: we handle this by including the paper's final register-reversal SWAP layer in the circuit. The *observed* success at $m = n$ (≈ 0.577) always exceeds the exact-QFT lower bound $4/\pi^2$, and the observed success curve dominates the theoretical AQFT lower bound at every m by a wide margin. The degenerate $m = 1$ point (Hadamard) shows a spuriously high 0.75: this is the special case where $r|2^L$ so $c = 0$ is always "good" and Hadamard puts weight $1/2^{L/2} \cdot |S_0|/\sqrt{|S_0|}$ etc. on $c = 0$; the paper's LB is a worst-case bound and does *not* require this degenerate case to be tight. **Matches C3.**

4.4 Figure

See `report/evidence/figure_aqft_replication.png` for a two-panel plot: (left) state fidelity vs. m for $n \in \{4, 6, 8\}$; (right) period-finding success and paper's lower bound vs. m for $L \in \{6, 8\}$.

5 Verdict

REPLICATED. Three of the paper's four in-scope claims (C1, C2, C3, C5) are reproduced on real Qiskit statevector simulation; the fourth (C4, decoherence trade-off) is scoped out of this replication — it requires a specific noise model to be well-defined — and is captured as an open question below. No fabricated numbers; all outputs are in `report/evidence/` as JSON.

Open Questions

See `report/open_questions.json` for the machine-readable form.

- **Q1.** Under what specific decoherence model does $AQFT_m$ with $m < n$ actually *beat* exact QFT end-to-end (Sec. 5 of the paper)? The paper's argument is heuristic ($O(Lm)$ gates vs. $O(L^2)$ gates times per-gate error); a modern re-analysis under, e.g., depolarizing noise with $p = 10^{-3}$ per two-qubit gate on $L = 8$ would pin down the optimal m^*.

- **Q2.** Our $m = 1$ (Hadamard) success prob 0.75 is an artifact of $r \mid 2^L$: how much of the paper's "AQFT-1 already useful" narrative survives when $r \nmid 2^L$ (e.g. $N = 21$, $r = 6$, $L = 9$)?

- **Q3.** We tested Haar-random pure states for fidelity; the paper's periodicity-estimation bound is for the specific periodic-comb state. What is the *worst-case* input state for $AQFT_m$ fidelity, and does it match the average behavior we saw?

- **Q4.** The gate-count formula $L + (2L - m)(m - 1)/2$ suggests sublinear-in-L savings for fixed m. In a fault-tolerant surface-code setting where each controlled-T-gate costs a magic-state distillation budget, does the AQFT actually reduce the *logical resource count*, or is it dominated by Hadamards / SWAPs?

- **Q5.** The paper's phase-difference bound $|\varepsilon| \leq 2\pi L/2^m$ is quite loose in our measurements (observed $\max |\varepsilon|$ is often $\leq 0.1\times$ the bound). Is there a tighter analytic bound of the form $C(n, m)$ that our data supports empirically?